LAB EXPERIMENT RECORD

24-10-2026---Set 30: Mobile App Usage Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	BANDLAPALLI BHANUTEJA REDDY
Register Number	192325016
Date	24 OCTOBER 2026
Day	SATURDAY

Aim

To analyze mobile app usage data in R and Tableau using a histogram with density plot, a scatter plot with trend line, a bar chart and an interactive dashboard.

Dataset: Mobile_App_Usage.csv

User ID	Gender	Age	Screen Time (hrs)	App Usage Count	Data Used (GB)	Satisfaction	Usage Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

Question 1: Histogram and Density Plot of Screen Time

Question: Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

Aim: To study how daily screen time is distributed among users using a coloured histogram and density plot.

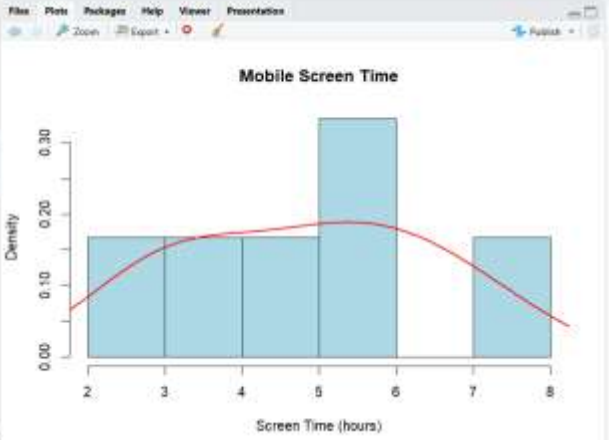
```
# PRANEL APALLI BHARGAVA REDDY
mobile <- data.frame(
  ID = c("U1", "U2", "U3", "U4", "U5", "U6"),
  Gender = c("Male", "Female", "Male",
             "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Screen_Time = c(4.5, 6.0, 3.2, 7.1, 2.8, 5.4),
  App_Count = c(18, 25, 12, 30, 10, 22),
  Data_Used = c(2.4, 3.8, 1.6, 4.5, 1.2, 3.1),
  Satisfaction = c(3, 5, 3, 5, 2, 4)
)

print(mobile)
# SET 30 - Q1

hist(mobile$Screen_Time,
     probability = TRUE,
     breaks = 5,
     col = "lightblue",
     main = "Mobile Screen Time",
     xlab = "Screen Time (hours)")

lines(density(mobile$Screen_Time),
      col = "red",
      lwd = 2)
```

```
2 U2 Female  22        6.0       25       3.8           5
3 U3   Male  19        3.2       12       1.6           3
4 U4 Female  21        7.1       30       4.5           5
5 U5   Male  23        2.8       10       1.2           2
6 U6 Female  20        5.4       22       3.1           4
> # SET 30 - Q1
>
> hist(mobile$Screen_Time,
+      probability = TRUE,
+      breaks = 5,
+      col = "lightblue",
+      main = "Mobile Screen Time",
+      xlab = "Screen Time (hours)")
>
> lines(density(mobile$Screen_Time),
+       col = "red",
+       lwd = 2)
```

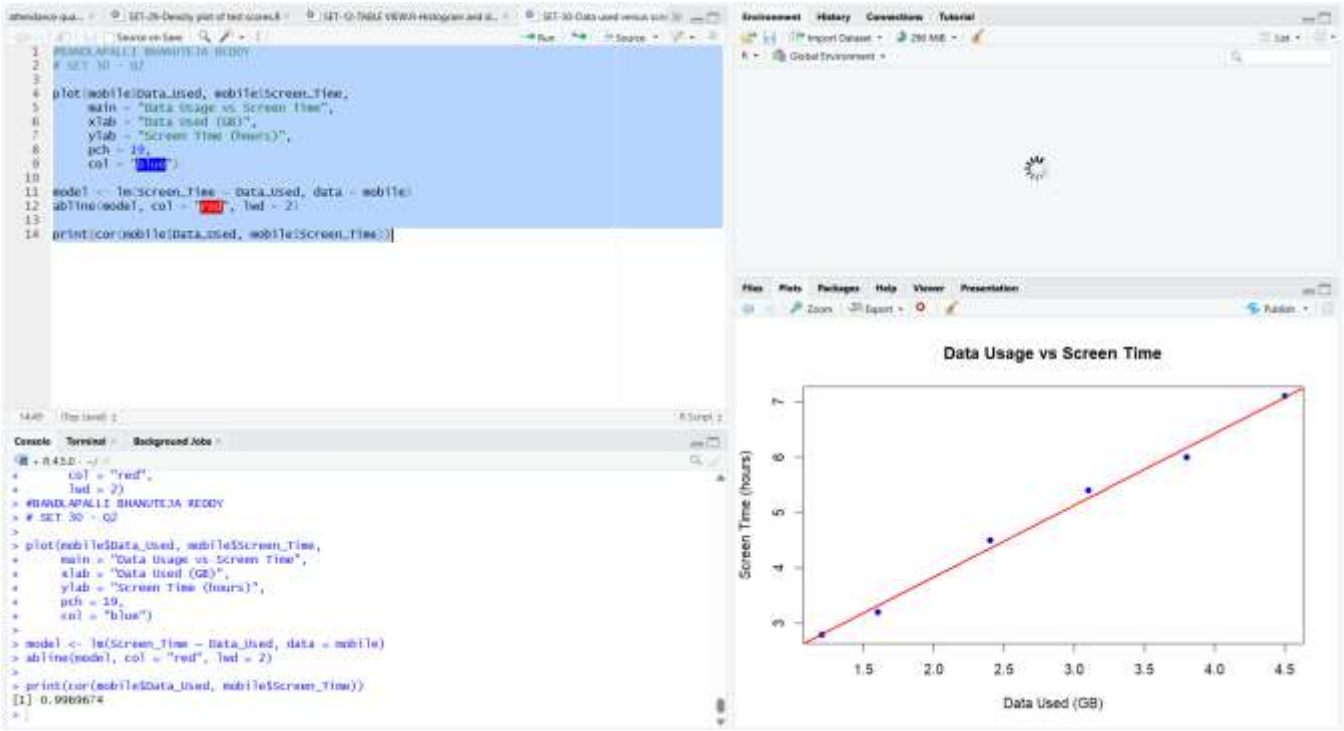


Result: Screen time ranges from 2.8 to 7.1 hours with a mean of about 4.8 and a median of about 5.0, and the users are spread widely with no single sharp peak.

Question 2: Scatter Plot of Data Used vs Screen Time

Question: Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

Aim: To explore how data usage and screen time are related using a scatter plot with a trend line and a correlation value.

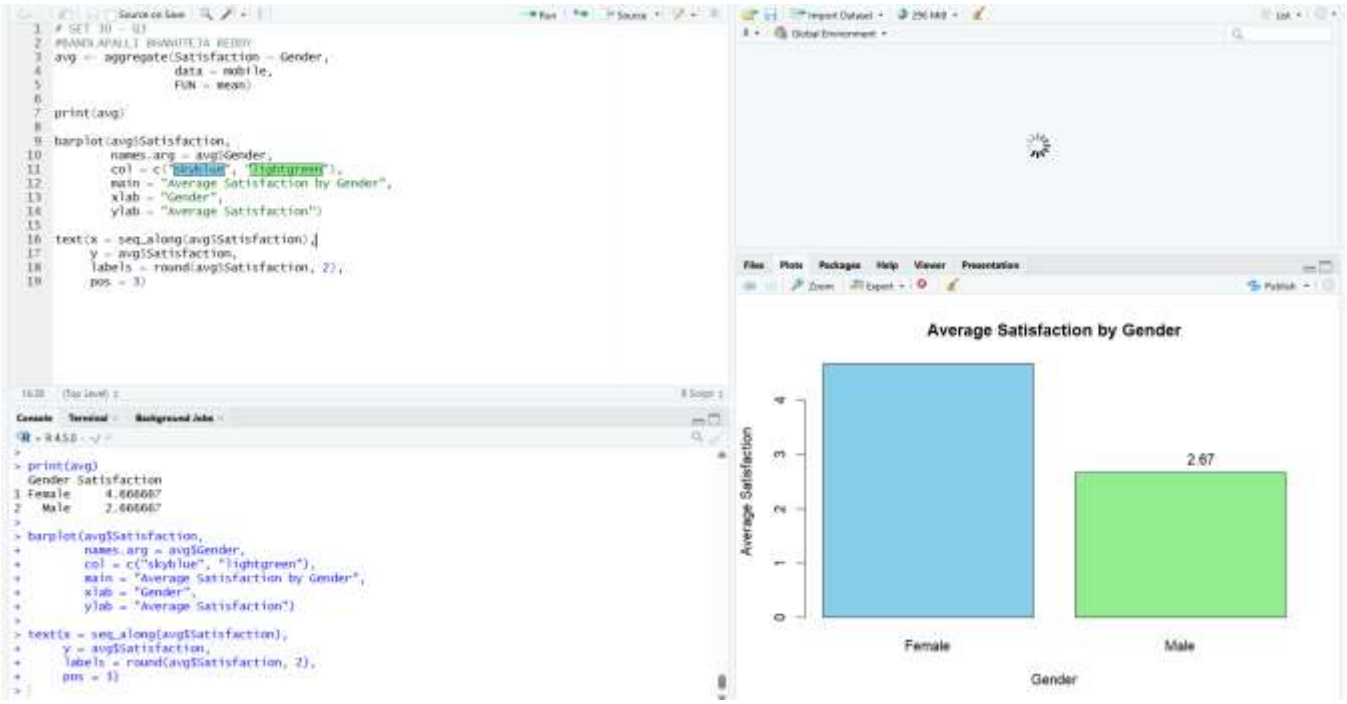


Result: The correlation is about 0.997, a very strong positive relationship: each extra hour of screen time goes with roughly 0.77 GB more data, so longer use means more data.

Question 3: Average Satisfaction by Gender

Question: Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

Aim: To find the average satisfaction score for each gender and show it in a bar chart with value labels.



Result: Female users average 4.67 and male users 2.67, so female users are clearly more satisfied, and they are also the heavier users of the apps.

Question 4: Interactive Tableau Dashboard

Question: Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.

Aim: To build a mini interactive Tableau dashboard with a bubble chart, a usage trend line chart and filters for Gender and Usage_Date.



Result: The bubble chart shows female users in the upper right with bigger bubbles, the line chart shows average screen time falling from 5.25 hours in January to 5.15 in February and 4.1 in March, and the filters update both views.

Overall Result

Observation: The average screen time is about 4.8 hours, the average data used is about 2.8 GB and the average satisfaction is 3.7, with heavier users being more satisfied.